ADINA SHAIKH (DOUBLE GOLD MEDALIST)

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EDUCATION

Master of Science (MSc) in Advanced Design of Ships and Offshore Structures (EMship+)

Erasmus Mundus Joint Master's Degree – Funded by the European Commission *Sep 2025-Oct 2027*

- Université "Dunărea de Jos" de Galați, Romania (Year 1) | Consortium of 7 European Universities (Belgium, France, Germany, Portugal, Spain, Romania)

National University of Sciences And Technology Karachi, Pakistan

Bachelor of Mechanical Engineering **(CGPA: 3.97)** *Sep 2021 – May 2025*

- Achievements: President's Gold Medal, Rector's Gold Medal, Awarded Sindh Government merit-based scholarship, Rector's Scholarship (x7)

PROFESSIONAL EXPERIENCE

Intern – National Pingtung University, Taiwan Pingtung, Taiwan

International Internship Pilot Programme (IIPP), NSTC *June 2025- August 2025*

Project: Design and Modeling of Vanadium Flow Redox Battery (VFRB)

- Designed and modeled a **single-cell VFRB** with end plates, current collectors, flow frames (serpentine channels), graphite electrodes, Ion membrane, and sealing gaskets using **SolidWorks and Siemens NX**.
- Extended the design to a **three-cell stack**, incorporating bipolar plates, tie rods, and redesigned gaskets for modular scalability.
- Developed **exploded and assembled CAD models** for single-cell and multi-cell systems
- Contributed to advancing research in **renewable energy storage technologies** through scalable battery system design.

Turkish Aerospace Industries Inc Ankara, Turkey

Structural Design Intern *July 2024 – Aug 2024*

- Developed expertise in **Geometric Dimensioning and Tolerancing (GD&T)**, applying it to ensure precision in design specifications and compliance with aerospace industry standards.
- Developed a strong foundation in **aerospace design methodologies**, incorporating best practices in structural design for aircraft components.
  Gained in-depth exposure to the **manufacturing and assembly processes** of HÜRKUŞ 2, HÜRJET, and Airbus components, building a comprehensive understanding of the **end-to-end aerospace production lifecycle**.
- Worked closely with cross-functional teams in **design, manufacturing,** and **safety**, applying engineering solutions to real-world aerospace challenges.

Fauji Fertilizer Company Limited (Plant I & II) Goth Machi, Sadiqabad, Pakistan

Plant Maintenance Engineering Intern *June 2024 – July 2024*

- Gained in-depth experience in **plant maintenance**, focusing on the mechanical systems crucial to fertilizer production, including **10+** centrifugal pumps, **8** compressors, **15+** valves, and **6** heat exchangers.
- Conducted a detailed analysis of **Pump P-102**, enhancing its performance by **15%** through problem-solving, fault detection, and implementing maintenance best practices.
- Collaborated closely with senior engineers on maintenance strategies for critical equipment, including over **500 meters of piping systems** and their fittings, tailored to the demands of continuous fertilizer production.
- Participated in **5** plant inspections and **3** site visits, gaining practical exposure to large-scale industrial processes specific to fertilizer production, contributing to improved safety and efficiency measures.

PUBLICATIONS

Research Article: Shaikh, A., & Dawood, M. (2023). Estimating the Tidal Energy Potential within the Indus Deltaic Creeks, Focusing on Techno-Economic Challenges in Pakistan. https://doi.org/10.2139/ssrn.4515087

PROJECTS

Multiphysics Analysis and Design Optimization of Composite Nose Radome for Supersonic Combat Aircraft

- Developed an optimized radome design for **AESA radar systems** integrated **Frequency Selective Surface (FSS)** technology focusing on reducing drag, managing thermal and mechanical stresses, and improving electromagnetic transmission for supersonic combat applications.
- Conducted thermal, electromagnetic, aerodynamic, and structural analyses, ensuring resilience in **high-stress flight conditions** with temperatures exceeding 350°C.
- Collaborated with **Turkish Aerospace Industries** to incorporate industry insights and ensure applicability in high-speed.

Mathematical Modelling of Solid Oxide Fuel Cell

- Proposed a multigeneration system employing a solid oxide fuel cell powered by an anaerobic digester fed with animal manure using MATLAB & Simulink, achieving 73% efficiency in utilizing methane for transport fuel.

HVAC Load Calculations

- Researched and developed an excel sheet to calculate the total heat load of a space based on external and internal heat gain

Designed and Manufactured a Prosthetic Hand

- Researched and manufactured a single actuator prosthetic hand, which can perform precision and power grip.

SKILLS

Interpersonal: Problem Solving, Communication, Leadership, Time Management, Adaptability

Technical: SolidWorks | Siemens NX | ANSYS | MATLAB | Simulink | Finite Element Analysis (FEA) | Python | MS Office